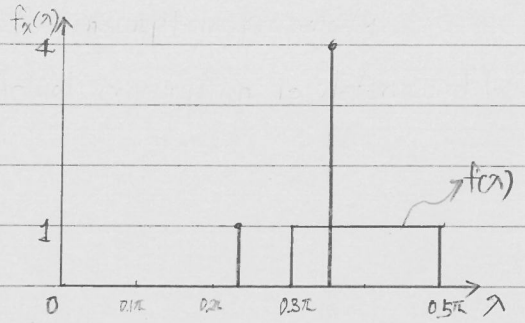
MA0367 Time Series Analysis and Forecasting HW#3

2. $x_k = \sin(4\pi k/17) + 2\sin(7\pi k/19)$,

$f(\lambda) = 1_{[0.3\pi, 0.5\pi]}(\lambda)$, $\lambda \in (0, \pi)$

sol) $f_x(\lambda) = \delta_{4\pi/17}(\lambda) + 2^2 \delta_{7\pi/19}(\lambda)$



$4\pi/17 \notin [0.3\pi, 0.5\pi]$

$7\pi/19 \in [0.3\pi, 0.5\pi]$

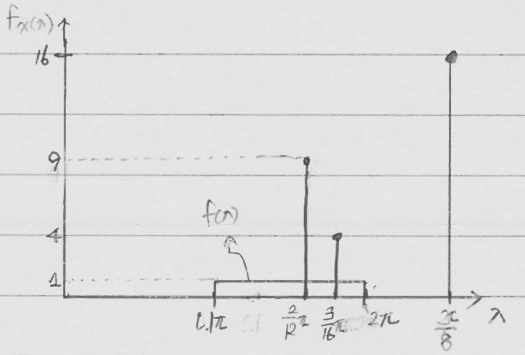
$f_y(\lambda) = f_x(\lambda) \cdot f(\lambda) = 4\delta_{7\pi/19}(\lambda)$

$\therefore y_k = 2\sin(7\pi k/19)$

3. $x_k = 4\sin(2\pi k/8) + 3\sin(2\pi k/12) + 2\sin(3\pi k/16)$

$f(\lambda) = 1_{[0.1\pi, 0.2\pi]}(\lambda)$, $\lambda \in (0, \pi)$

sol) $f_x(\lambda) = 4^2 \delta_{2\pi/8}(\lambda) + 3^2 \delta_{2\pi/12}(\lambda) + 2^2 \delta_{3\pi/16}(\lambda)$



$2\pi/8 \notin [0.1\pi, 0.2\pi]$

$2\pi/12 \in [0.1\pi, 0.2\pi]$

$3\pi/16 \in [0.1\pi, 0.2\pi]$

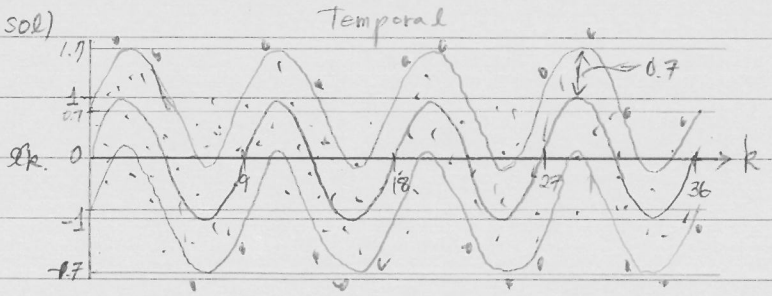
$f_y(\lambda) = f_x(\lambda) \cdot f(\lambda) = 9\delta_{2\pi/12}(\lambda) + 4\delta_{3\pi/16}(\lambda)$

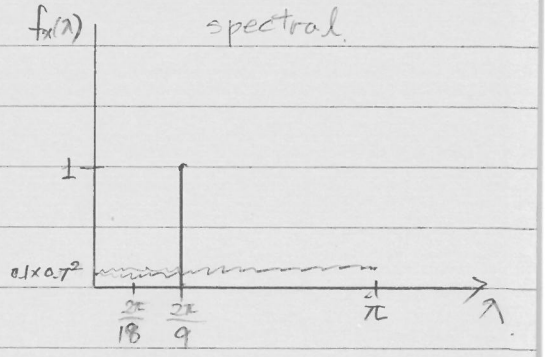
$\therefore y_k = 3\sin(2\pi k/12) + 2\sin(3\pi k/16)$

5. $x_k = \sin(2\pi k/9) + 0.7\epsilon_t$

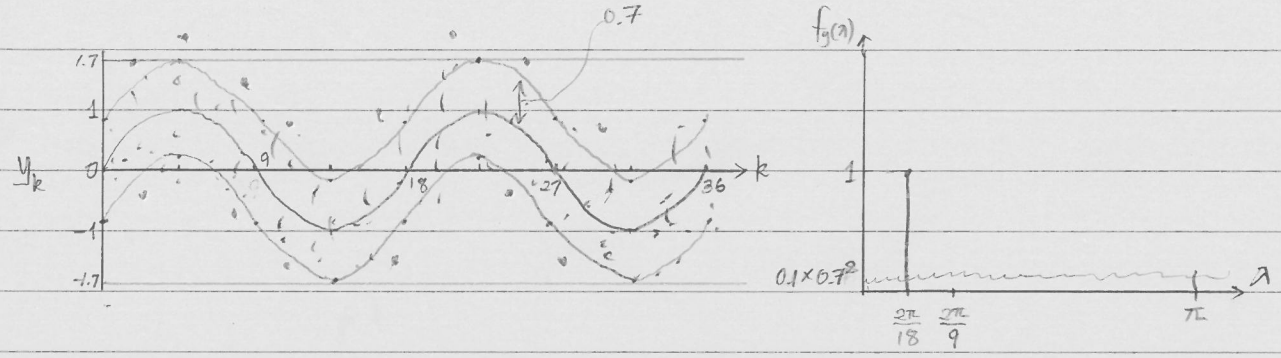
$y_k = \sin(2\pi k/18) + 0.7\epsilon_t$

sol)





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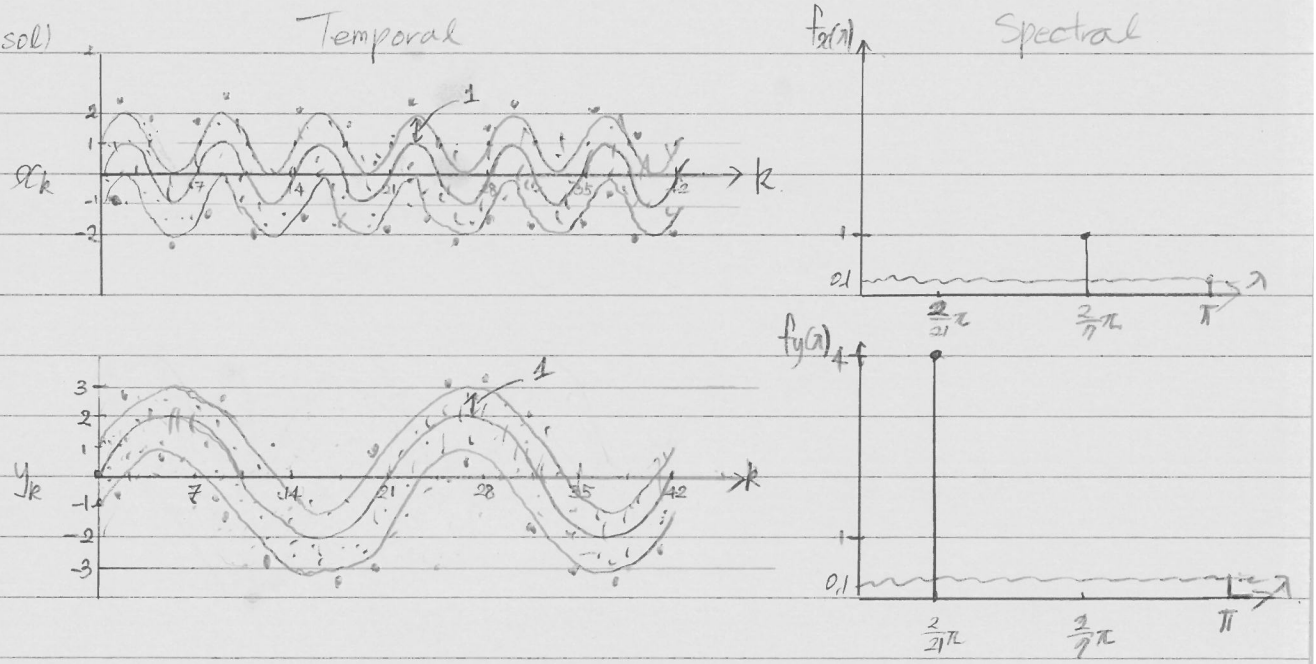
The time series x_k and y_k have the same level of noise.

The periodogram for x_k has the peak at $\frac{2\pi}{9}$ and its amplitude is 1. However, the periodogram for y_k has the peak at $\frac{2\pi}{18}$ and its amplitud is 1.

6. $x_k = \sin(2\pi k/7) + \epsilon_k$

$y_k = 2\sin(2\pi k/21) + \epsilon_k$

sol)



The time series x_k, and y_k have the same level of noise.

The periodogram for x_k has the peak at $\frac{2\pi}{7}$ and its amplitude is 1. However, the peridogram for y_k has the peak at $\frac{2\pi}{21}$ and its amplitude is 4.